

# EC2 Linux Web Server Deployment

## Apache HTTPD — Beginner-Friendly Command Explanation

This guide explains how to create an Amazon EC2 Linux instance and host a simple webpage using Apache HTTPD.

---

### 1. Create an EC2 Instance

1. Sign in to the **AWS Management Console**.
2. Open **EC2** and click **Launch instance**.
3. Enter a name, for example: `apache-web-server`.
4. Choose an **Amazon Linux** operating system.
5. Select a small instance type for testing, such as `t2.micro` or `t3.micro`.
6. Select or create a **key pair** for SSH access.
7. Ensure the instance receives a **Public IPv4 address**.
8. Launch the instance and wait until its status is **Running**.

### Security Group Rules

Configure the security group with these inbound rules:

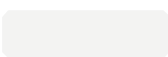
| Type                                                                                      | Port | Purpose                        |
|-------------------------------------------------------------------------------------------|------|--------------------------------|
|  SSH   | 22   | Connect to the server remotely |
|  HTTP  | 80   | Allow normal website traffic   |
|  HTTPS | 443  | Allow secure website traffic   |

 For better security, allow SSH port 22 only from your own IP address.

---

### 2. Connect to the Linux

Use SSH from your computer:

-  `-i /path/to/key.pem` identifies your private key.
-  `ec2-user` is the usual Amazon Linux username.
-  `PUBLIC_IP_ADDRESS` is the public IPv4 address of the EC2 instance.
- 

### 3. Switch to the Root User

```
sudo su
```

This changes the current user to **root**. Root has full permission to install software and modify system files.

The command prompt normally changes from `$` to `#`.

### 4. Update Linux Packages

```
yum update -y
```

-  `yum` is the package manager used by many Amazon Linux systems.
-  `update` downloads and installs available package updates.
-  `-y` automatically confirms the installation prompts.

### ☐ 5. Install Apache HTTPD

```
yum install httpd -y
```

This installs **Apache HTTP Server**. On Amazon Linux, the Apache package is called `httpd`.

---

## 6. Move to the Website Directory

```
cd /var/www/html
```

This moves into Apache's default website directory, called the **document root**.

Apache serves webpage files placed in:

```
/var/www/html
```

## 7. Create the HTML File

```
touch index.html
```

This creates an empty file named `index.html`.

Apache normally displays `index.html` as the default webpage.

---

## 8. Edit the HTML File with vi

```
vi index.html
```

This opens the file in the `vi` text editor.



## Save the Webpage

1. Press `i` to enter **insert mode**.

2. Type or paste your HTML code.
3. Press `Esc` to return to normal mode.
4. Type:

```
:wq
```

1. Press **Enter**.

### Meaning of `:wq`

- `:w` means **write/save** the file.
- `q` means **quit** the editor.
- Together, `:wq` saves the file and exits `vi`.

## ► 9. Start Apache HTTPD

```
service httpd start
```

This starts the Apache web server. Apache can now serve files from `/var/www/html`.

## 10. Start Apache Automatically After Reboot

```
chkconfig httpd on
```

This configures Apache to start automatically whenever the EC2 instance is restarted.

**i** On newer Linux systems, the modern equivalent commands are:

```
> systemctl start httpd
> systemctl enable httpd
> ```

-

# 🌐 11. Open the Website

In a browser, open:

```text
http: /PUBLIC_IP_ADDRESS/
```

Replace `PUBLIC_IP_ADDRESS` with the public IPv4 address of your EC2 instance.

## ✓ Expected Result

Your HTML page should appear in the browser.

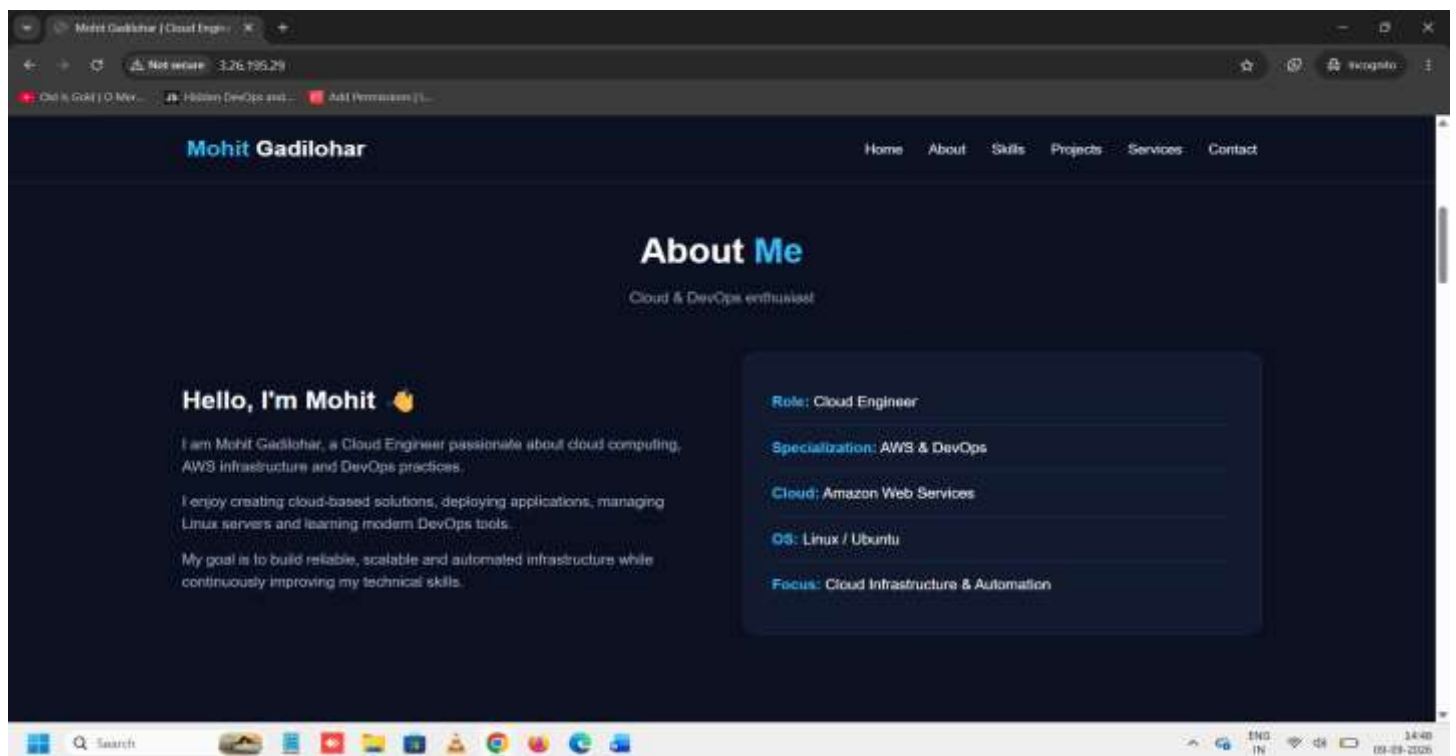
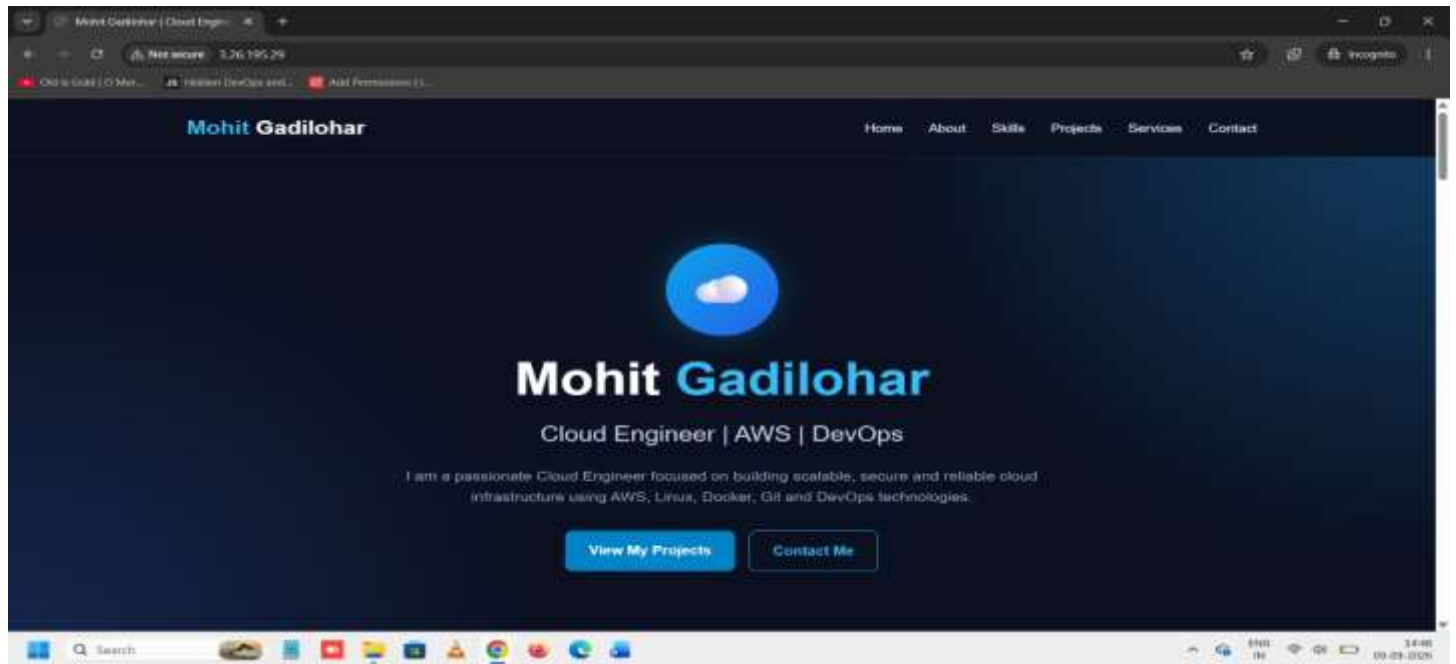
## 🔧 Command Summary

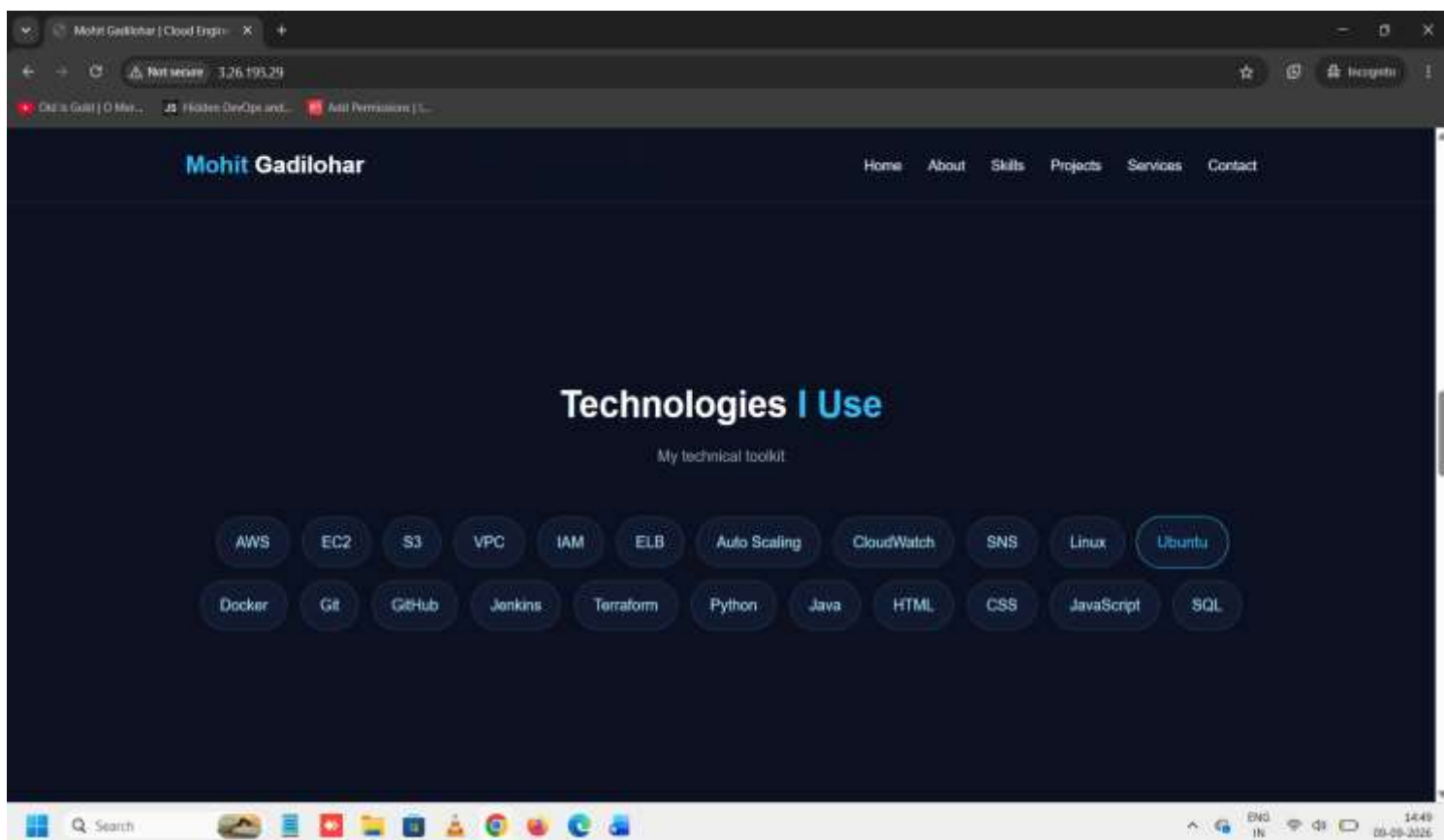
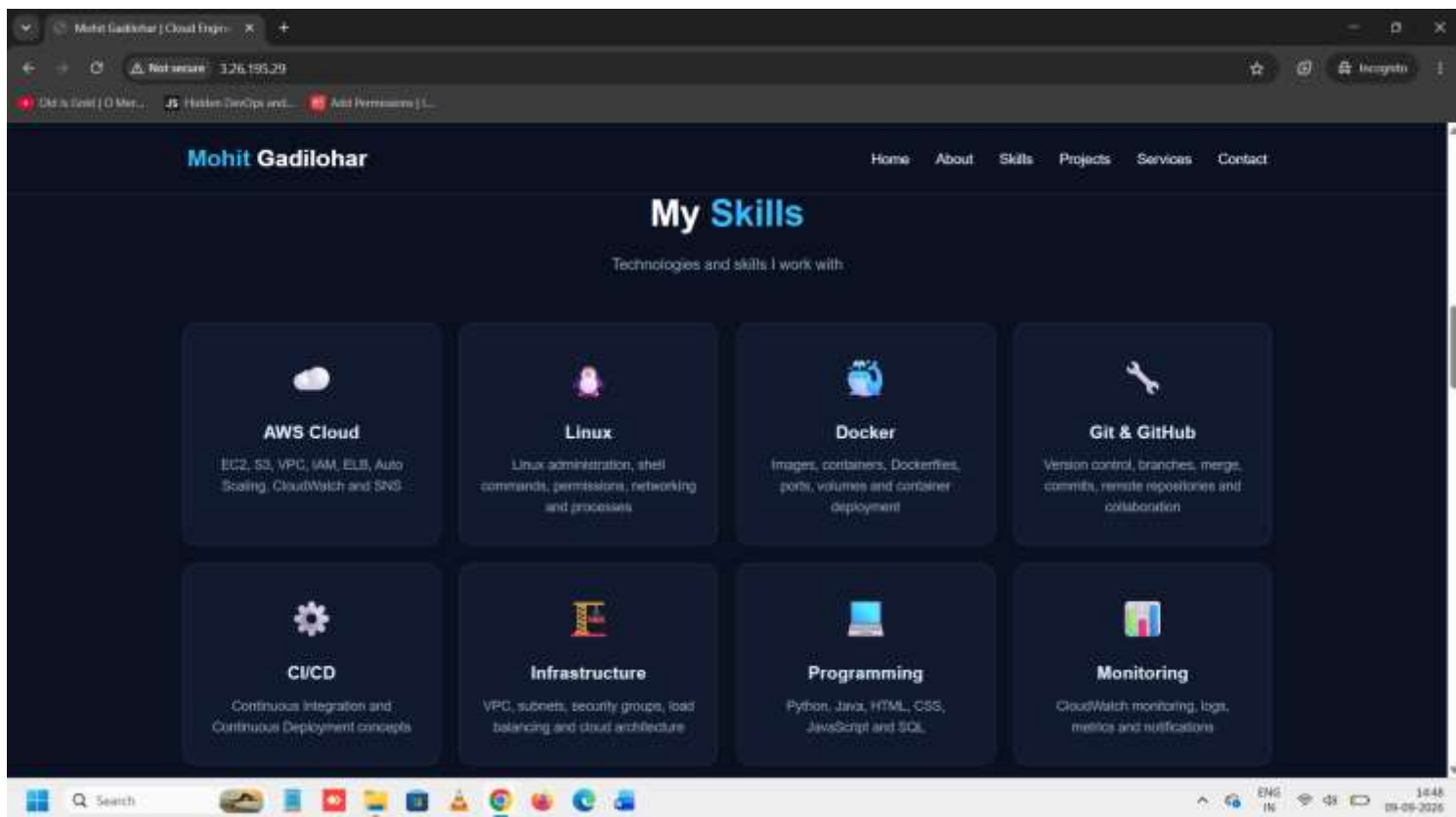
```
sudo su
yum update -y
yum install httpd -y
cd /var/www/html
touch index.html
vi index.html
:wq
service httpd start
chkconfig httpd on
```

🔧 These commands install Apache, create a webpage, start the web server, and configure it to start automatically after a reboot.

Copy the Instance public IP Address and paste the Incognito Mode

✔Show The Output





## My Projects

Practical cloud and DevOps projects

### PROJECT 01

#### AWS Static Website Hosting

Hosted a static website using Amazon S3. Configured bucket settings, static website hosting, permissions and website deployment.

[AWS S3](#) [HTML](#) [CSS](#) [Cloud](#)

### PROJECT 02

#### EC2 Linux Server

Provisioned an Ubuntu EC2 instance and configured security groups. Connected to the server using SSH/PUTTY and performed Linux administration tasks.

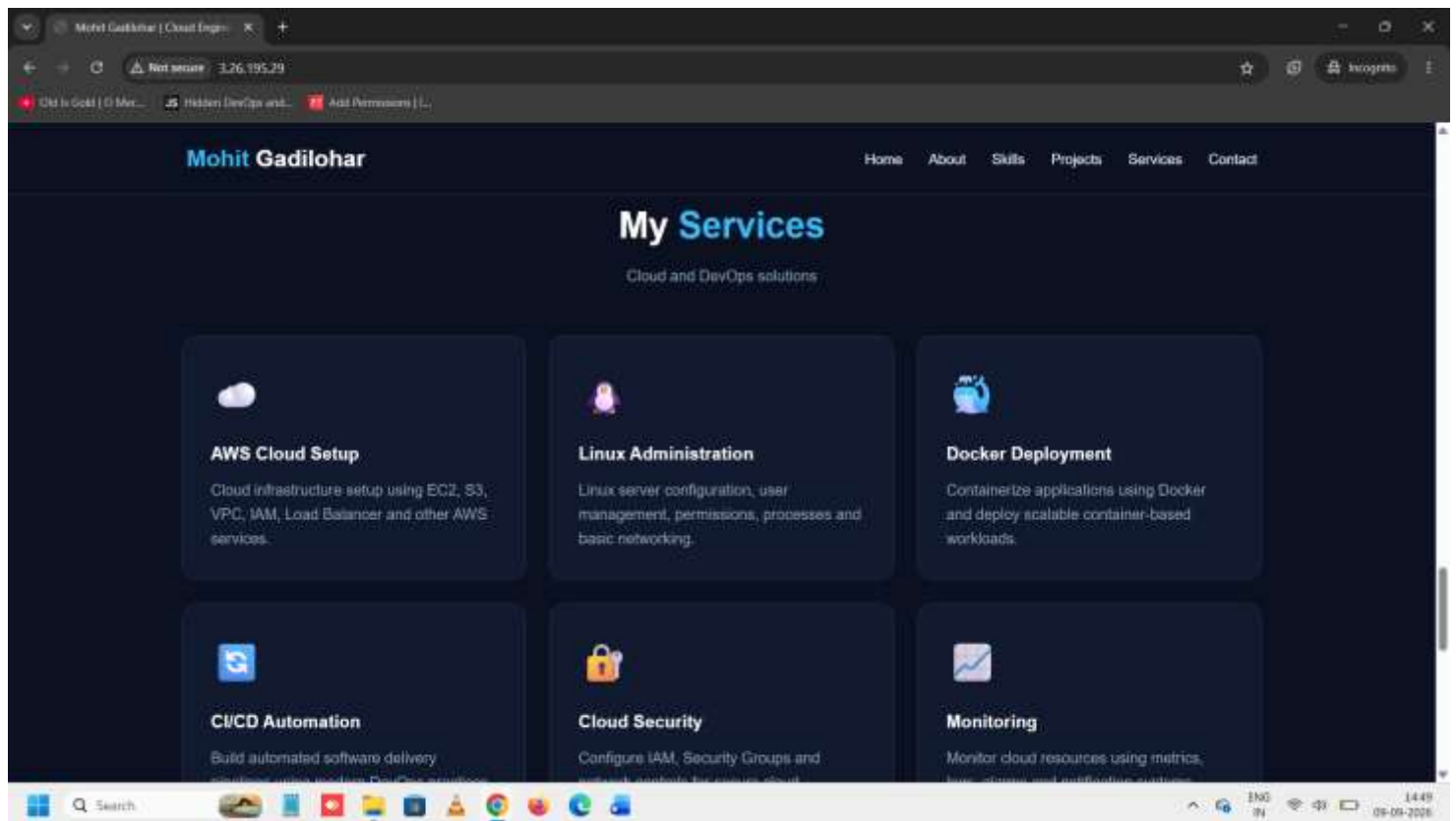
[AWS EC2](#) [Ubuntu](#) [Linux](#) [SSH](#)

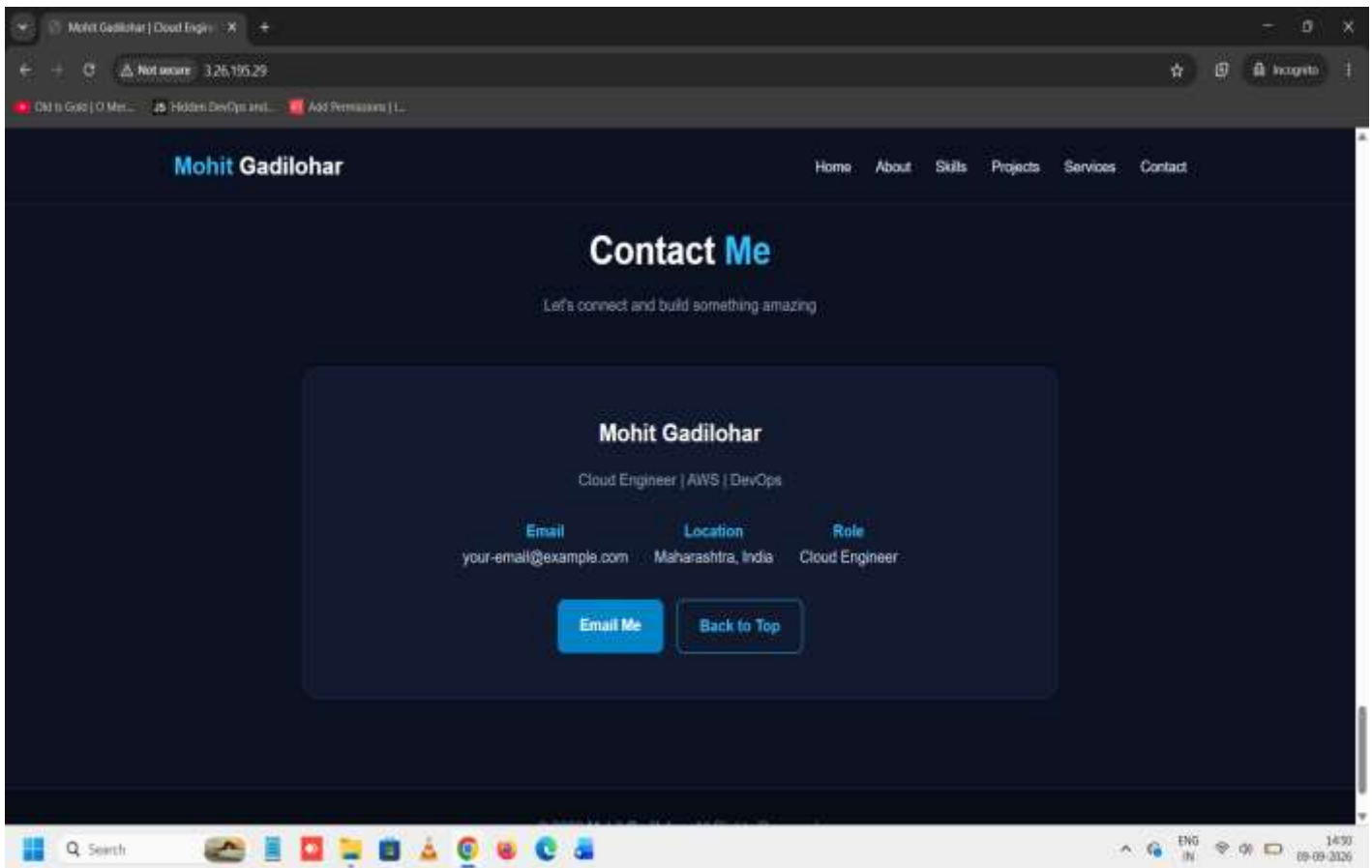
### PROJECT 03

#### AWS Load Balancer

Designed a load-balanced application environment using EC2 instances, Target Groups and an Application Load Balancer.

[ALB](#) [EC2](#) [Target Group](#) [AWS](#)





## About

**EC2 Linux with Apache (HTTPD) Web Server Deployment** is a cloud project that demonstrates hosting a website on an AWS EC2 Linux instance using the Apache web server. It covers EC2 setup, Linux configuration, security groups, Apache installation, and website deployment.

## Conclusion

The project provided practical experience in **AWS EC2, Linux administration, Apache web server, networking, and web deployment**, establishing a strong foundation for Cloud and DevOps engineering.